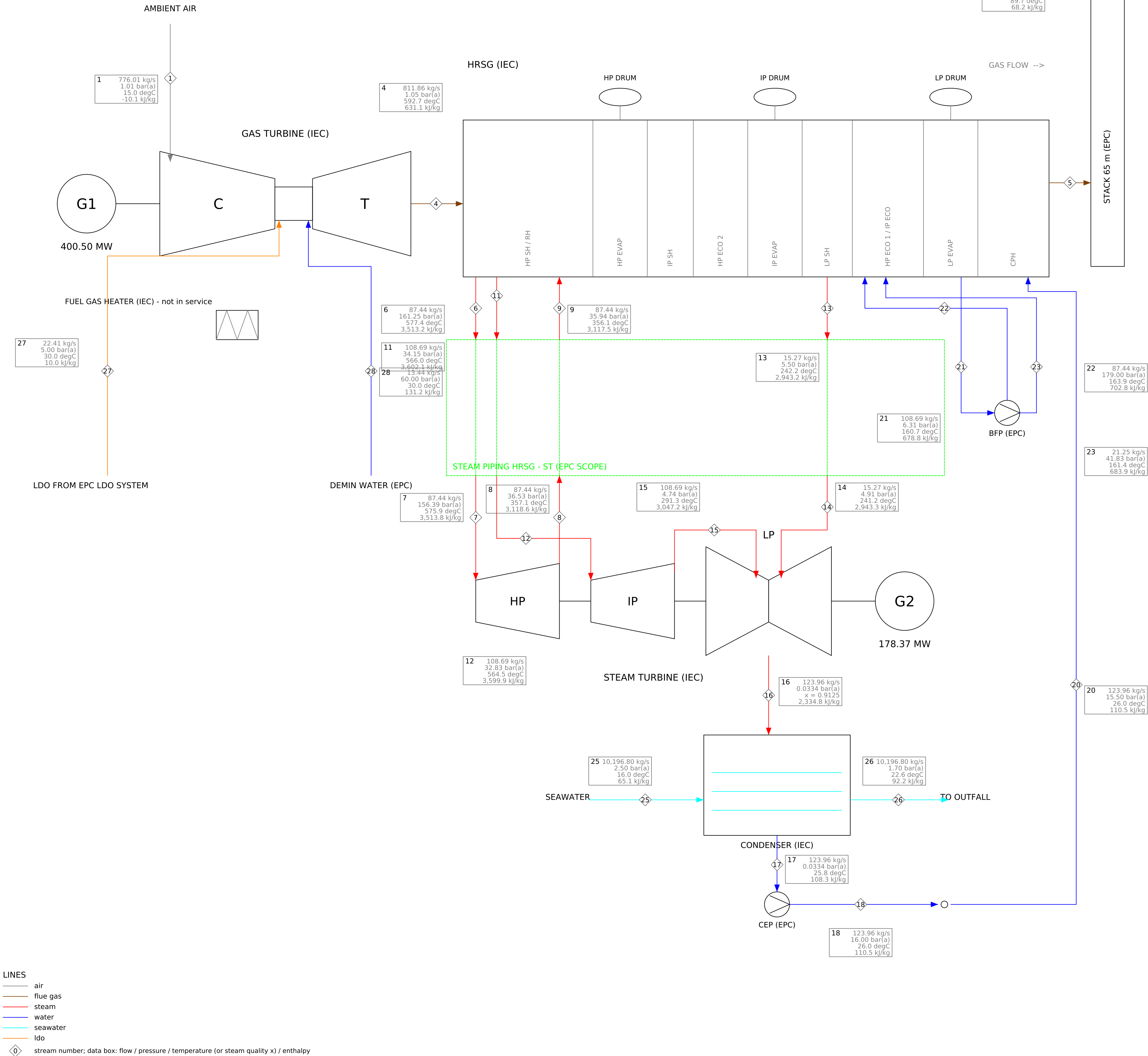




STREAM TABLE - SRC-LDO-100

No	Stream	kg/s	bar(a)	degC / x	kJ/kg	MW*
1	Ambient air to GT compressor inlet	776.01	1.01	15.0	-10.1	-7.8
4	GT exhaust gas to HRSG inlet	811.86	1.05	592.7	631.1	512.4
5	Flue gas at HRSG outlet to stack	811.86	1.01	89.7	68.2	55.4
6	HP steam at HRSG outlet (HP SH outlet)	87.44	161.25	577.4	3,513.2	307.2
7	HP steam at ST main stop valve inlet	87.44	156.39	575.9	3,513.8	307.2
8	Cold reheat steam at HP turbine exhaust	87.44	36.53	357.1	3,118.6	272.7
9	Cold reheat steam at HRSG reheater inlet	87.44	35.94	356.1	3,117.5	272.6
10	IP superheated steam (joins cold reheat inside the HRSG)	21.25	37.26	312.6	3,004.7	63.9
11	Hot reheat steam at HRSG outlet	108.69	34.15	566.0	3,602.1	391.5
12	Hot reheat steam at ST IP stop/control valve inlet	108.69	32.83	564.5	3,599.9	391.3
13	LP steam at HRSG outlet	15.27	5.50	242.2	2,943.2	44.9
14	LP steam at ST LP admission	15.27	4.91	241.2	2,943.3	44.9
15	IP turbine exhaust (crossover, internal)	108.69	4.74	291.3	3,047.2	331.2
16	LP turbine exhaust to condenser (UEEP)	123.96	0.0334	x 0.913	2,334.8	289.4
17	Condensate at hotwell outlet / CEP suction	123.96	0.0334	25.8	108.3	13.4
18	Condensate at CEP discharge	123.96	16.00	26.0	110.5	13.7
20	Condensate to HRSG condensate preheater inlet	123.96	15.50	26.0	110.5	13.7
21	Feedwater from LP drum to boiler feed pumps	108.69	6.31	160.7	678.8	73.8
22	HP feedwater at BFP discharge to HP economiser	87.44	179.00	163.9	702.8	61.5
23	IP feedwater at BFP interstage to IP economiser	21.25	41.83	161.4	683.9	14.5
25	Circulating seawater to condenser inlet	10,196.80	2.50	16.0	65.1	663.9
26	Circulating seawater from condenser outlet	10,196.80	1.70	22.6	92.2	939.9
27	LDO at liquid fuel skid inlet (IEC terminal point IF-02)	22.41	5.00	30.0	10.0	0.2
28	Demineralised water for NOx water injection (IF-03)	13.44	60.00	30.0	131.2	0.8

* energy flow m x h (reference: IAPWS-IF97 for water/steam, 25 degC sensible for gases, cp x T for seawater)

PLANT POWER BALANCE

GT output (generator terminals)	400.50	MW
ST output (generator terminals)	178.37	MW
Gross output	578.87	MW
Auxiliary loads	-9.35	MW
Transformer losses (GSU, UAT, MV/LV)	-1.73	MW
NET OUTPUT (HV terminals)	567.80	MW
Heat input (LHV)	961.22	MW
Gross heat rate (LHV)	5,978	kJ/kWh
NET HEAT RATE (LHV)	6,094	kJ/kWh
Net efficiency (LHV)	59.07	%
Fuel LHV (ISO 6976, from composition)	42.90	MJ/kg
Circulating water flow	36,112	m3/h
GT load	100.00	%
PLANT LOAD (NET, % OF SRC-NG-100)	92.48	%
Water injection (NOx)	13.44	kg/s

CHECKS (guarantees, limits, balances)

OK	energy balance CMX: residual +0.001 MW (tolerance 0.25 MW)
OK	energy balance COND: residual +0.047 MW (tolerance 0.95 MW)
OK	energy balance GT: residual +0.016 MW (tolerance 0.92 MW)
OK	energy balance HRSG: residual +1.860 MW vs stated casing loss 1.828 MW (tolerance 0.87 MW)
OK	energy balance PIPE: residual +0.286 MW (tolerance 1.02 MW)
OK	energy balance ST: residual -0.002 MW (tolerance 0.74 MW)
OK	net export 567.8 MW vs DP-OWNR-0051 connection capacity 640 MW (this case)
OK	gross output 578.87 MW vs IG-09 >= 572.00 (margin +1.20 %)
OK	gross heat rate 5,977.82 kJ/kWh vs IG-10 <= 6,020.00 (margin +0.70 %)
OK	net output 567.80 MW vs PG-15 >= 560.00 (margin +1.39 %)
OK	net heat rate 6,094.42 kJ/kWh vs PG-16 <= 6,150.00 (margin +0.90 %)
OK	guarantee cover: at IG-09/IG-10 limits and this HMB's auxiliaries the net plant is 560.9 MW / 6,139 kJ/kWh vs Owner PG-15 560 MW / PG-16 6,150 kJ/kWh

PROJECT	Alpilla 600 MW Combined Cycle Power Plant
TITLE	HEAT AND MASS BALANCE - SRC-LDO-100
CASE	Site Reference Conditions, LDO with water injection, base load (guarantee PG-15/
DOCUMENT	ALP-HMB-SRC-LDO-100
STATUS	preliminary - generated by engine hmb v1.1.1
DATA	b6f4b950c7f3b7e0 2026-09-29
NOTE	all values calculated from the database; do not edit